🧠 AI Fundamentals & Prompting for TPMs

Week 1 • Day 1 • Customer-Centric Foundations

Technical Product Manager Academy

👋 About Me

John Kidd

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- 20+ years building commercial software

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- Have working access to a generative AI assistant (ChatGPT, Claude, or equivalent)

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🎯 Where This Fits

Adjacent skills you'll touch

- Prompt engineering & AI-augmented research
- System design & architecture trade-offs
- Stakeholder mapping & negotiation
- Agile delivery and ADO mastery

Where to go next

- Capstone artifact carries into your real product work
- Ongoing TPM mentorship inside Expeditors
- Skills/SkillIQ post-academy assessments

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- Use an on-screen timer for breaks and lab timeboxes

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- Ask questions any time — chat, raised hand, or unmute

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In ~30 seconds each, please share:

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- A fun fact — one sentence

Recording & Notetaker Policy

Notetaker apps (Read AI, Otter, Fireflies, etc.) are not permitted in this session.

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Thanks for your cooperation.

🚀 Let's Get Started!

Day 1 — AI Fundamentals & Prompting

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- Detect and recover from hallucination, grounding failures, and false confidence
- Build a personal **Prompt Pattern Library** with your cohort triad

🕑 Today's Shape

Clock	Block	Outcome
09:00 – 09:15	Opening	Meet your triad, ground the week
09:15 – 10:30	Block 1 + Activity 1	What LLMs actually are
10:45 – 12:00	Block 2 + Activity 2	Four prompt patterns
13:00 – 14:15	Block 3 + Activity 3	Three TPM uses of AI
14:30 – 16:00	Block 4 + Activity 4 + Readouts	Verification & trust

Block 1 — What an LLM Actually Is

Enough mental model to stop being surprised by it

🔎 The Only Mental Model You Need

An LLM is a **very large statistical model of language** that predicts the next token given everything before it. It is not a database. It is not a reasoner. It does not "know" things.

✅ It is good at

- Rephrasing, summarizing, re-formatting
- Pattern-completing from strong context
- Drafting first passes quickly
- Stress-testing your own writing

❌ It is bad at

- Giving exact numbers or citations from memory
- Saying "I don't know"
- Reasoning about novel constraints without being shown them

🧠 Three Terms You Will Say Out Loud

Token

The unit the model reads and writes. Roughly ~4 characters of English. "Product" ≈ 1 token, "Dispatch" ≈ 1–2.

Context window

How many tokens the model can see in one call. Everything outside this window is invisible to it.

Temperature

A knob from deterministic (0) to creative (1+). PMs usually want low-to-mid for drafts, low for extraction.

PM implication: If the model can't see it in the prompt, it's guessing. Always ask yourself: *"What does this model know about my product right now? Nothing."*

👥 Activity 1 — "Same Question, Three Models"

Triads • 35 minutes

Each triad picks one open PM question (provided list of 10). Each member asks the *same* question to a different tool (ChatGPT, Claude, Gemini/Copilot).

- Compare: Where do the answers agree? Diverge? Hallucinate?
- Score each answer on **usefulness**, **verifiability**, **risk if wrong**
- Prepare a 60-second readout: "We would trust X for Y, and never trust it for Z"

⏱️ 10 min prompt · 15 min compare · 10 min write readout

Block 2 — Four Prompt Patterns

The smallest scaffold that meaningfully improves output

📚 The RCCF Scaffold

Every time you ask an AI for non-trivial PM help, include four things:

Role, Context, Constraints, Format.

Slot	Question it answers	PM example
Role	Who is talking?	"Act as a senior technical product manager reviewing a PRD"
Context	What does the model need to see?	User persona, the problem, prior attempts
Constraints	What must be true of the answer?	"Flag any claim that lacks evidence"
Format	What shape should it take?	"Markdown table with columns X, Y, Z"

📄 Before & After

❌ Weak prompt

```
Write me a user persona for a dispatch app.
```

Output is generic, confident, un-grounded, un-actionable.

✅ Scaffolded prompt

```
Role: Senior TPM for a field-service SaaS.
Context: Target user = dispatcher at an HVAC
company, 50–200 techs, 10+ yrs industry, low
tolerance for screens. Pain: re-routing mid-day
when a tech calls out sick.
Constraints: Persona must be grounded in the
behaviors above; flag any field that is an
assumption I'd need to validate.
Format: Markdown persona sheet with a final
"Assumptions to validate" section.
```

🔄 Prompting is a Dialogue, Not a Command

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- **First pass:** RCCF scaffold

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- **First pass:** RCCF scaffold
- **Second pass:** "Critique the above as if you were a skeptical engineer"

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- **First pass:** RCCF scaffold
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- **Third pass:** "Now rewrite it, addressing those critiques"

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- **First pass:** RCCF scaffold
- **Second pass:** "Critique the above as if you were a skeptical engineer"
- **Third pass:** "Now rewrite it, addressing those critiques"
- **Always:** "What in this answer are you least confident about?"

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👥 Activity 2 — Rewrite This Prompt

Triads • 40 minutes

Each triad receives three weak prompts taken from the FieldPulse backlog. Rewrite each one using RCCF, then run the original and the rewrite side-by-side.

- Capture: What changed in the output? What did *not* change?
- Add one prompt to your triad's **Prompt Pattern Library**
- Star the prompt your triad would actually reuse on Monday morning

⏱️ 15 min rewrite · 15 min run · 10 min library entry

Block 3 — Three Recurring TPM Uses

Where AI actually earns its seat at the table

🔍 Use 1 — Accelerated Research

The job: Quickly build context about an unfamiliar domain, user group, or competitor landscape — enough to ask smart questions, not to ship decisions.

✅ Good fit

- "Explain HVAC dispatcher workflows to me in the language they'd use"
- "What are five things I might be naive about here?"
- "Give me five search queries that will find primary-source material"

❌ Bad fit

- "What is the market size of dispatch software?" (will hallucinate)
- "Who is the leader?" (outdated and confident)
- "Cite a study that proves…" (fabricated citations)

📝 Use 2 — Summarization with Evidence Tags

When you feed a model a transcript, survey, or support-ticket batch, ask it to **tag each claim with where in the source it came from**. Unsourced claims are the ones to interrogate.

```
Role: Research analyst.
Context: Below are 14 support tickets from FieldPulse dispatchers.
Task: Extract the top 5 recurring frustrations.
Constraints:
- Each frustration must cite 2+ ticket numbers as evidence
- Flag any frustration you inferred without direct evidence
Format: Markdown list; each entry: Frustration / Evidence tickets / Confidence (H/M/L).
```

🧿 Use 3 — Self-Critique Loop

Before sharing a PRD, persona, or problem statement, run a structured critique pass:

1. "Read the following as a skeptical engineer. What three questions will you ask me first?"
2. "Read the following as the stakeholder who most often blocks launches here. What objection worries me most?"
3. "Read the following as the customer in the persona. Where does it not ring true?"

> **Why it works:** You get three perspectives in five minutes without socializing a half-baked draft with real people.

👥 Activity 3 — The Three Hats

Triads • 40 minutes

Each triad takes the FieldPulse one-paragraph problem brief (provided).

Using AI, run it through three critique hats: **Engineer**, **Blocker-stakeholder**, **Target customer**.

- Capture the single most uncomfortable question each hat raised
- Decide which one your triad cannot answer yet — that becomes your Day 3 research question

⏱️ 10 min each hat · 10 min synthesize

Block 4 — Verification & Trust

How grown-up TPMs use AI without being fooled by it

⚠️ The Four Failure Modes

Failure	What it looks like	Mitigation
Hallucination	Confident fabricated facts, citations, APIs	Treat outputs as leads, not sources. Verify numerics & names manually.
Context starvation	Plausible but generic answer	Add Role + Context explicitly. Paste in the source material.
Sycophancy	Agrees with whatever premise you give	Ask for counter-arguments and rank them by strength.
Anchoring	Your framing shapes its output	Ask for the opposite position as a second turn.

🔢 Evidence Tiers (we'll use this all week)

When you write something down — persona traits, pain points, problem statements — tag each claim with its evidence tier so reviewers can calibrate trust.

Tier	Source	Trust
Observed	You watched it happen, or recorded data shows it	High
Reported	A user said it in an interview or ticket	Medium
Inferred	You deduced it from related evidence	Low
AI-generated	A model produced it — not yet validated	Lowest; must be upgraded before use

👥 Activity 4 — Prompt Pattern Library (Publish)

Triads • 45 minutes • Readouts

Each triad finalizes a shared **Prompt Pattern Library** for use the rest of the academy. Minimum four entries:

- One **Research** prompt
- One **Summarization-with-evidence** prompt
- One **Critique-hat** prompt
- One **"Where could you be wrong?"** prompt

⏱️ 25 min draft · 10 min test · 10 min triad readout

🎉 Day 1 Wrap

What we built

- A mental model of what an LLM is (and isn't)
- The RCCF prompt scaffold
- Three reusable TPM uses: research, summarization, critique
- A Prompt Pattern Library per triad

What opens tomorrow

- **Working Backwards** — we write the press release *before* the product
- Your triad will use today's prompts to draft and critique your first PR/FAQ

Bring to Day 2: Your triad's Prompt Pattern Library + one domain you'd like to draft a Working-Backwards PR for.

💬 Questions & Reflection

Before you leave: what's the one prompt you'll use tomorrow morning?